

# Macroeconomic Effects of UI Extensions at Short and Long Durations

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# Introduction

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# Motivation

- Do UI extensions prolong periods of high unemployment?
  - Regular UI benefits last a maximum of 26 weeks in most states.
- GE forces could *amplify, weaken, or overturn* the PE effects.
  - Longer UI reduces the incentives of firms to post vacancies.
  - UI extensions transfer resources to high MPC households.
- Empirical challenges in estimating the GE effects.
  - Reverse causality: Automatic triggers and new UI programs.

# What does this paper do?

New approach based on the features of Extended Benefits program.

- The program provide additional weeks of benefits during periods of high unemployment in a state.
- All states are subject to mandatory “trigger rules” for extensions, but they may also adopt **optional** ones.
- Compare states that qualify for the same rules but have adopted different ones.
- **Identification assumption:** Historical option adoption is orthogonal to current economic conditions.

# Headline results

A 13-week extension...

- raises  $u$  by 0.28 pp. when baseline duration  $< 60$  weeks.
- has no effect on  $u$  when baseline duration  $\geq 60$  weeks.

Macro estimate at short durations  $\approx$  the low end of micro estimates.

Calibration of search-and-matching model with endogenous search effort implies small GE effects of UI extensions.

## Rules for UI Extensions

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# The Extended Benefits Program

- EB is a federal program that has since 1970 provided additional weeks of UI when certain conditions are met.
- A mix of mandatory and **optional** ones, which yield the variation in potential benefit duration across states.
- IUR: insured unemployment rate; TUR: total unemployment rate.

# Optional rules

Three optional trigger rules in effect.

- “IUR option”: If 13-week MA of the state IUR  $> 6\%$ .
  - “TUR option.”: If 3-month MA of the state TUR  $> 6.5\%$  &  $> 110\%$  of the max of 3-month MA of TUR 1 or 2 year prior.
    - A second tier that triggers 7 additional weeks: If TUR  $> 8\%$  &  $> 110\%$  of the 3-month MA of the state TUR 1- or 2-years prior.
  - The extensions from “IUR option” and “TUR option” do NOT cumulate, i.e., they only make a 13-week extension more likely.
- ⇒ Most optional EB extensions are 13 weeks or 20 weeks.

# Trigger rules

All states must adopt the mandatory rule since 1982.

- 1) A 13-week MA of state IUR  $> 5\%$ , and
  - 2) The current 13-week MA of the IUR  $> 120\%$  of the average of the 13-week MA of the IUR one and two years prior.
- Once an EB condition triggers on, the resulting UI extension remains active for at least 13 weeks.
- If a trigger rule continues to be satisfied for longer than 13 weeks, the UI extension remains on until all triggers lapse.
- Once all EB conditions lapse, UI extensions must remain off for at least 13 weeks.

# Example: Arkansas

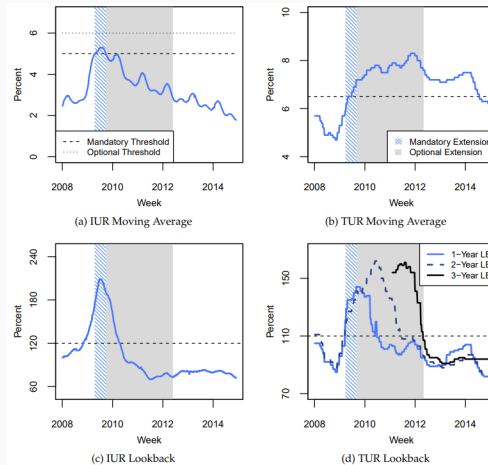


Figure 1: Trigger Variables in Arkansas during and after the Great Recession

- The potential benefit duration could have increased by 20 weeks.

| Rule Type                        | Rule Description                                                            | Effective Years      |
|----------------------------------|-----------------------------------------------------------------------------|----------------------|
| 13 Weeks                         |                                                                             |                      |
| Mandatory                        | IUR MA $\geq$ 4% and IUR Lookback $\geq$ 120%                               | 1970–1971, 1981–1982 |
| Mandatory                        | (IUR MA $\geq$ 4% and IUR Lookback $\geq$ 120%) or National IUR $\geq$ 4.5% | 1972–1981            |
| Optional                         | IUR MA $\geq$ 5%                                                            | 1976–1982            |
| Mandatory                        | IUR MA $\geq$ 5% and IUR Lookback $\geq$ 120%                               | 1982–                |
| Optional                         | IUR MA $\geq$ 6%                                                            | 1982–                |
| Optional                         | IUR MA $\geq$ 5% and 3-year IUR Lookback $\geq$ 120%                        | 2011–2013            |
| Optional                         | TUR MA $\geq$ 6.5% and 1- or 2-year TUR Lookback $\geq$ 110%                | 1993–                |
| Optional                         | TUR MA $\geq$ 6.5% and 1-, 2-, or 3-year TUR Lookback $\geq$ 110%           | 2011–2013            |
| 7 Additional Weeks               |                                                                             |                      |
| Optional                         | TUR MA $\geq$ 8.0% and 1- or 2-year TUR Lookback $\geq$ 110%                | 1993–                |
| Optional                         | TUR MA $\geq$ 8.0% and 1-, 2-, or 3-year TUR Lookback $\geq$ 110%           | 2011–2013            |
| Interactions with other Programs |                                                                             |                      |
| Optional*                        | FSB only triggered if EB triggered                                          | 1975–1978            |
| Optional*                        | States that recently triggered EB also triggered FSC early on               | 1982–1983            |
| Optional*                        | Triggering EB also triggered TEUC benefits                                  | 2002–2004            |

Figure 2: Trigger Rules for the Extended Benefits Program

# Changes in Option Status

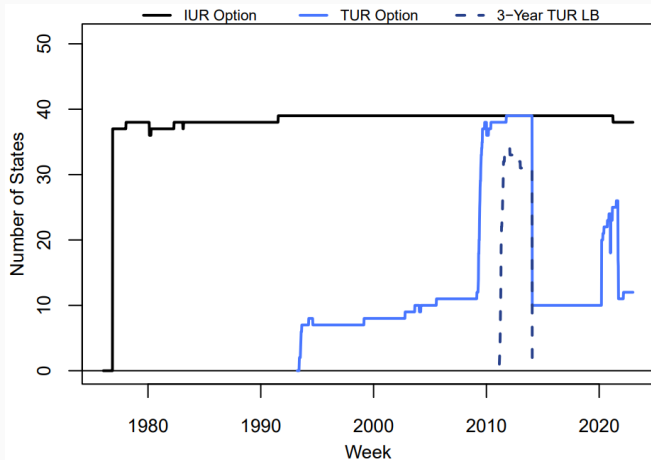


Figure 3: Changes in Option Status

## Changes in Option Status

| Description              | Optional Rules |     |            |       |         |
|--------------------------|----------------|-----|------------|-------|---------|
|                          | TUR            | IUR | TUR 3-Y LB | Total | Percent |
| Federal Funding          | 104            | 0   | 68         | 172   | 85      |
| <i>ARRA (2009)</i>       | 51             | 0   | 68         | 119   | 59      |
| <i>General (2009+)</i>   | 22             | 0   | 0          | 22    | 11      |
| <i>Fam. First (2020)</i> | 29             | 0   | 0          | 29    | 14      |
| <i>TEUC (2003)</i>       | 2              | 0   | 0          | 2     | 1       |
| Reagan Reforms           | 0              | 6   | 0          | 6     | 3       |
| Option Creation          | 7              | 0   | 0          | 7     | 3       |
| Discretionary            | 8              | 9   | 0          | 17    | 8       |
| Total                    | 119            | 15  | 68         | 202   | 100     |

Figure 4: Sources of Changes in Trigger Rules

- The program is 50% federally funded and has at times (2009 - 2013) been made 100% federally funded.

# A threat to identification strategy

- Why do some states adopt the options when they are “free” while others do not?
- Differing views about the extensions & political attitudes.
  - 10 of the 12 states<sup>1</sup> that did not adopt the TUR option in 2012 had below-median Obama vote shares in the 2008 election.
- Perhaps states experiencing worse economic conditions are more likely to opt-in.

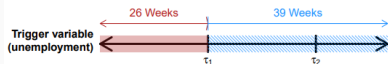
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<sup>1</sup>Wyoming, Oklahoma, Utah, Arkansas, Louisiana, Nebraska, Mississippi, North Dakota, South Dakota, Montana, Iowa, and Hawaii

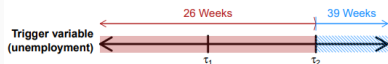
# Empirical Methodology

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## State with option on

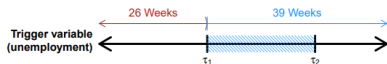


## State with option off

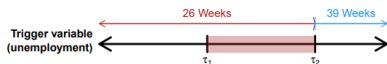


■ Treatment ■ Control

## State with option on



## State with option off



■ Treatment ■ Control

Figure 5: Illustration of Identification

- Two thresholds:  $\tau_1$ , an optional one, and  $\tau_2$ , a mandatory one.
- Include separate fixed effects for each of these three intervals.
- The variation in benefits within the interval is NOT due to differences in  $u$ , but only due to the options.

# Risk sets

|                      | IUR Option | TUR Option | 3-Year TUR Lookback Option |
|----------------------|------------|------------|----------------------------|
| 13-Week EB Tier      | ✓          | ✓          | ✓                          |
| 20-Week EB Tier      |            | ✓          | ✓                          |
| FSC "Reachback" Tier | ✓          |            |                            |
| FSC Tier II          | ✓          |            |                            |
| TEUC "X" Tier        |            | ✓          |                            |

**Figure 6:** Determinants of Risk Sets

- Partition the state space into "risk sets" based on which options a state satisfies the trigger rules for.
- A check-mark indicates that this option can trigger this tier at some point during our sample period.
- The risk sets at a given point in time are all the different combinations of cells with check-marks at that point in time.

# Regressions Conditioning on Risk Sets

$$y_{s,t+h} = \beta_h \widehat{W}_{s,t} + \text{qual. controls}_{h,s,t} + \delta_{h,s} + \Gamma' x_{h,s,t} + \epsilon_{h,s,t}$$

$$\text{qual. controls}_{h,s,t} = \sum_{z,t} \alpha_{z,h,t} I_s(z, t)$$

- $s$  denotes states,  $t$  is time measured in weeks
- $y_{s,t+h}$  is an outcome of interest in week  $t + h$
- $\widehat{W}_{s,t}$  is the treatment variable of interest
- $\delta_{h,s}$  is a set of state fixed effects
- $x_{h,s,t}$  is a set of additional controls

# Regressions Conditioning on Risk Sets

$$y_{s,t+h} = \beta_h \widehat{W}_{s,t} + \text{qual. controls}_{h,s,t} + \delta_{h,s} + \Gamma' x_{h,s,t} + \epsilon_{h,s,t}$$

$$\text{qual. controls}_{h,s,t} = \sum_{z,t} \alpha_{z,h,t} l_s(z, t)$$

- $\widehat{W}_{s,t}$ : difference between the actual and the **counterfactual** duration if state  $s$  at time  $t$  had adopted no options.
- EB program is the only program with optional trigger rules.
- Differences in the duration of regular benefits / the incidence of other programs across  $s$  do NOT contribute to variation in  $\widehat{W}_{s,t}$ .

# Regressions Conditioning on Risk Sets

$$y_{s,t+h} = \beta_h \widehat{W}_{s,t} + \text{qual. controls}_{h,s,t} + \delta_{h,s} + \Gamma' x_{h,s,t} + \epsilon_{h,s,t}$$

$$\text{qual. controls}_{h,s,t} = \sum_{z,t} \alpha_{z,h,t} l_s(z, t)$$

- $l_s(z, t)$  is equal to one if at time  $t$ , state  $s$  is in risk set  $z$ .
- Dummy variables for each  $z$  interacted with time fixed effects.
- *Within*  $z$ , the variation in  $\widehat{W}_{s,t}$  arises from option adoption ONLY.

The methodology builds on two major data collection efforts.

- “UI Benefits Calculator”
  - predicts whether a state triggers onto federal UI extensions as F (state-level trigger variables, EB program options adopted).
  - yields **counterfactual** potential benefit durations for each state had the state made a different choice regarding options.
- A dataset on real-time values of the “trigger variables”
  - enter into the trigger rules of these federal programs.

# Sources

A case-by-case analysis of federal and state UI legislation based on

- The Chronology of Federal Unemployment Compensation Laws,
- news reports,
- notices in the Federal Register,
- memoranda written by the DOL to state UI agencies, and
- correspondence with members of the Division of Fiscal & Actuarial Services at the DOL.

Collection and digitization of trigger variables and option statuses

- The DOL's weekly Trigger Notices
- archives from the Division of Fiscal & Actuarial Services at DOL
- The Library of Congress, several university libraries, and the University of Texas Libraries.

# UI Benefits Calculator

| Program          | Applicable State-Weeks | Incorrect | % Correct |
|------------------|------------------------|-----------|-----------|
| EB (1976–2022)   | 125,001                | 52        | 99.96     |
| FSB (1976–1978)  | 5,457                  | 112       | 97.95     |
| FSC (1982–1985)  | 6,732                  | 14        | 99.79     |
| EUC (1991–1994)  | 5,814                  | 0         | 100.00    |
| TEUC (2002–2004) | 5,508                  | 0         | 100.00    |
| EUC (2008–2013)  | 14,535                 | 40        | 99.72     |
| Total            | 163,047                | 218       | 99.87     |

Figure 7: UI Benefits Calculator Performance

- Given real-time measures of trigger variables and option adoptions, the calculator returns an accurate trigger status.
- Some mistakes in the Trigger Notices, i.e., the state did pay EB while the Trigger Notice indicated it did not.

# LAUS, CPS, CES, and Administrative UI Data

- Local Area Unemployment Statistics (LAUS) from the BLS: Unemployment, the labor force, and population.
- Current Population Survey (CPS) data: An alternative measure of unemployment.
- BLS' Current Employment Statistics (CES) program: Employment.
- DOL administrative data: UI payments, the number of initial claims, and the number of UI recipients by state.
  - Dollar amount of benefits deflated by the national PCE deflator.

## Recap of the identification strategy

- Two innovations: (i) carefully defined treatment variable, (ii) time - risk set fixed effects.
- Arguable better than naive approach with the level of actual duration as treatment and state & time fixed effects.
- There's NO way to directly test the identification assumption.
- One way: check for pre-trend differences.

# Endogeneity of Option Changes

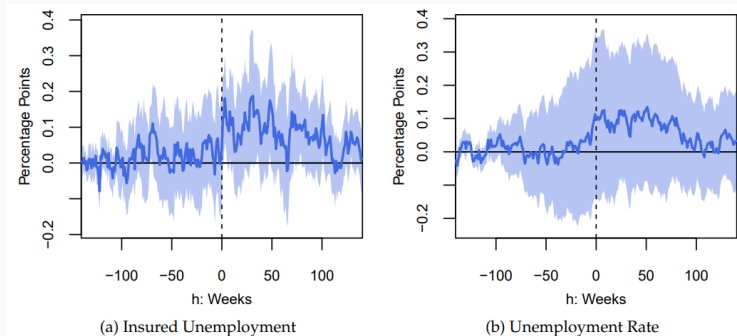


Figure 8: What Predicts Option Switches?

- Treatment: an indicator for whether a state has adopted any option in  $t$ .
- Alternative: Drop 2-years before and after any change in option status.

# Regressions Conditioning on Risk Sets

$$y_{s,t} = \beta W_{s,t} + \text{qual. controls}_{s,t} + \delta_s + \Gamma' x_{s,t} + \epsilon_{s,t}$$

$$y_{s,t} \equiv \sum_{\ell=-6}^6 y_{s,t+\ell}$$

- $y_{s,t}$  is the quarterly sum of  $y_{s,t}$  centered on week  $t$ .
- $W_{s,t}$  is the quarterly sum of potential benefit duration.
- $\beta$  is a sum. measure of the contemporaneous quarterly effect.
- Instrument for  $W_{s,t}$  with  $\hat{W}_{s,t}$ , estimate by 2SLS.

## Results

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# Short versus Long Duration

- In addition to the EB program, Congress has passed laws that have temporarily extended UI during each recession since 1958.
- They create (additional) variation in the “base” potential benefit duration over time.
- The long duration sample comes mostly from the Great Recession and its aftermath.

# The Short Baseline Potential Benefit Duration Sample

|                        | Unemployment    |                | Insured Unemployment |                |
|------------------------|-----------------|----------------|----------------------|----------------|
|                        | 1980–2019       | 1980–2022      | 1980–2019            | 1980–2022      |
| $W_{s,t}$ , Short Dur. | 0.28<br>(0.13)  | 0.44<br>(0.16) | 0.60<br>(0.08)       | 0.93<br>(0.19) |
| $W_{s,t}$ , Long Dur.  | -0.01<br>(0.10) | 0.05<br>(0.10) | 0.23<br>(0.12)       | 0.28<br>(0.13) |
| Observations           | 103494          | 111298         | 103494               | 111298         |

Figure 9: Effects of a 13-Week Benefit Extension

- 2SLS estimates of  $\beta$ .
- The effect on IUR is about twice as large as the effect on TUR.
- Some people who would otherwise exhaust benefits may collect UI for longer even if they are not unemployed for longer.

# The Short Baseline Potential Benefit Duration Sample

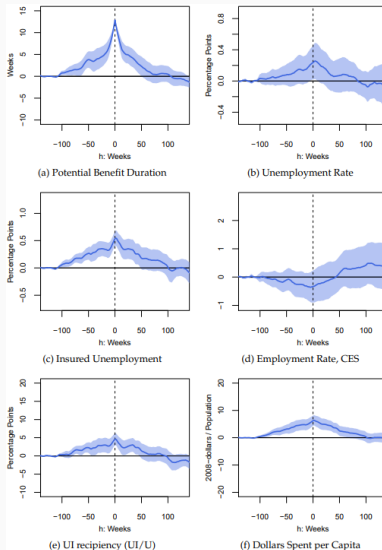


Figure 10: Short Baseline Potential Benefit Durations

# The Long Baseline Potential Benefit Duration Sample

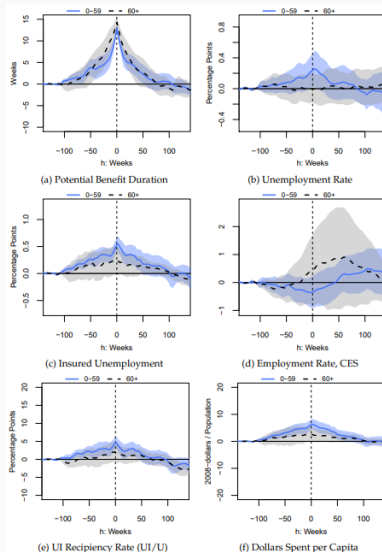


Figure 11: Long vs. Short Baseline Potential Benefit Durations

# Short versus Long Durations

UI extensions matter less at longer baseline durations.

- The number of workers impacted falls over time.
  - Over 1994-2021, median duration of spells = 11 wks. In 2012, 19 wks.
- Workers may anticipate that they are unlikely to use a UI extension far in the future.
- Those most at risk of long-term unemployment may also be less employable and thus respond less to UI extensions.
- UI extensions during the recoveries after deep recessions often involve political uncertainty around these programs.

## How Big Are the Estimates?

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# Relationship to Micro Estimates

- A plain-vanilla search model that abstracts from GE effects.
- The LOM for unemployment:  $U_t = U_{t-1} - f_t U_{t-1} + s E_{t-1}$ 
  - $f_t$  = job finding rate,  $s$  = constant separation rate.
- The steady state unemployment rate is  $u^* = s/(s + f)$
- Using  $d \log(u^*) = -(1 - u^*) d \log(f)$  and  $D = 1/f$ ,

$$\frac{d \log(D)}{d \log(\Pi)} = \frac{1}{1 - u^*} \frac{d \log(u^*)}{d \log(\Pi)}$$

- Schmieder and von Wachter (2016):  $\frac{d \log(D)}{d \log(\Pi)} \approx 0.33$  to  $0.41$ .

# Relationship to Micro Estimates

- Multiply the range by UI reciprocity rate (0.36)  $\Rightarrow$  0.12 to 0.15.
- Average  $u = 6.79\%$ , percentage change in  $u = 0.28/6.79$ .
- Average baseline potential benefit duration = 34 weeks, percentage change =  $13/34$ .
- The ratio is  $\frac{0.28}{0.07} \times \frac{34}{13} = 0.11$ . Divide by  $1 - u^*$  yields 0.12.

# Macro Model of UI Benefit Extensions

- Standard DMP with
  - (i) limited duration of UI benefits
  - (ii) endogenous search effort
  - (iii) incomplete take up of UI benefits
  - (iv) transitions in and out of the labor force.
- Time is discrete (monthly) and the discount factor is  $\beta$ .
- Firms post  $v$ . Workers are either ( $e$ ), ( $u$ ) or inactive ( $n$ ).
- Firm-worker matches produce output  $p$ , separated with  $\delta$ .
- Matching is random through a CRS matching function  $M(S, v)$

# Macro Model of UI Benefit Extensions

- The flow value of unemployment  $b(\tau) = \mathbb{1}[\tau > 0]b_{UI} + b_L$ ,

$\tau$  = the number of periods of UI benefits left.

- Unemployed workers exert search effort  $s$  at cost  $c(s)$ 
  - $c(s) = \kappa s^{1+\frac{1}{\gamma}}$ :  $c(0) = 0$ ,  $c'(s) > 0$  and  $c''(s) < 0$
  - They are matched to firms at rate  $s\lambda(\theta)$ .
- Optimal search effort  $s(\tau)$  depends on  $\tau$ .
- Aggregate search effort  $S = \sum_{\tau=0}^T u(\tau)s(\tau)$ .
  - $u(\tau)$  is the mass of unemployed with  $\tau$  periods of UI benefits left.

# Macro Model of UI Benefit Extensions

- At the beginning of an unemployment spell, unemployed workers draw an i.i.d. take-up cost  $\xi \sim G(\xi)$ .
- Each period workers draw a home production shock with probability  $\iota$ , which leads them to leave the labor force.
- They re-enter the labor force through  $u$  with probability  $\rho$ .
- Workers who exit the labor force lose eligibility to UI benefits. They requalify with probability  $h$  once they find a job.
- Nash Bargaining with the worker bargaining share  $\alpha$ .
- Free-entry conditions for vacancies.

| Symbol     | Parameter Description           | Value  | Source/Target                                              |
|------------|---------------------------------|--------|------------------------------------------------------------|
| $\beta$    | Discount factor                 | 0.996  | Annual interest rate of 5%                                 |
| $c$        | Vacancy posting cost            | 545.2  | $\theta = 1$                                               |
| $\mu$      | Matching efficiency             | 1.00   | Normalization                                              |
| $\eta$     | Elasticity of matching function | 0.72   | Shimer (2005)                                              |
| $\alpha$   | Worker's bargaining share       | 0.72   | Hosios condition                                           |
| $T$        | Potential UI benefit duration   | 10     | Average PBD prior to EB extensions                         |
| $b_{UI}$   | UI benefit                      | 1,061  | Average UI benefit in 2008 dollars                         |
| $p$        | Aggregate productivity          | 3,032  | $\frac{b_{UI}}{p} = 0.35$ ; Anderson and Meyer (1997)      |
| $b_L$      | Flow value of leisure           | 1,721  | $\frac{E(b_{UI}+b_L)}{p} = 0.71$ ; Hall and Milgrom (2008) |
| $b_N$      | Flow value of home production   | 3,032  | $b_N = p$                                                  |
| $h$        | Requalification probability     | 0.167  | Average requalification period of 6 months                 |
| $\delta$   | Separation rate                 | 0.0162 | EU transition rate                                         |
| $\iota$    | Home production shock           | 0.0031 | Unemployment rate                                          |
| $\rho$     | Home production shock           | 0.0060 | Labor force participation rate (LFPR)                      |
| $\mu_G$    | Mean of take-up costs           | 5,992  | Insured unemployment rate                                  |
| $\sigma_G$ | Std. of take-up costs           | 13,086 | Response of UI reciprocity rate to PBD                     |
| $\kappa$   | Search cost scaling parameter   | 27,331 | UE transition rate                                         |
| $\gamma$   | Search cost elasticity          | 0.47   | Micro-elasticity of duration to PBD                        |

Figure 12: Calibrated Parameter Values in the Model

| Variable                                                      | Target | Model |
|---------------------------------------------------------------|--------|-------|
| <i>A. Steady-state averages:</i>                              |        |       |
| Unemployment rate (%)                                         | 7.13   | 7.13  |
| Insured unemployment rate (%)                                 | 2.90   | 2.90  |
| Job-finding rate (%)                                          | 25.0   | 25.0  |
| Job-separation rate (%)                                       | 1.61   | 1.61  |
| Labor force participation rate (%)                            | 65.8   | 65.8  |
| <i>B. Steady-state responses to 3-month extension of PBD:</i> |        |       |
| Micro-elasticity of duration of UI recipients                 | 0.33   | 0.33  |
| Macro-response of UI reciprocity rate (p.p.)                  | 4.94   | 4.96  |

Figure 13: Targeted Moments

| Variable             | Data    |          | Model (Micro) |          | Model (Macro) |          |
|----------------------|---------|----------|---------------|----------|---------------|----------|
|                      | $T = 8$ | $T = 17$ | $T = 8$       | $T = 17$ | $T = 8$       | $T = 17$ |
| Unemployment         | 0.28    | -0.01    | 0.29          | 0.20     | 0.43          | 0.30     |
| Insured unemployment | 0.60    | 0.23     | 0.48          | 0.25     | 0.52          | 0.30     |

Figure 14: Responses to 3-Month UI Extension

- The micro effects hold labor market tightness constant.
- The macro effect is the full SS response including the GE effects that operate through labor market tightness.
- Most unemployed job seekers do not expect to stay unemployed long enough. This force is not strong enough to explain the data.

# Conclusion

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# Conclusion

- Impressive work on the data over the last 50 years.
  - Short vs. long baseline duration is new.
- Macroeconomic effects of UI extensions on the unemployment rate are minimal at times when UI durations are already long.
- The effects are substantial when initial UI benefit durations are initially short.
- The reason, however, is not clear.